

TRAPS: Therapeutic Response Analysis via Pathway-informed Stratification

Abstract

Cancer treatment planning requires decisions across multiple clinical dimensions at once. Clinicians must determine whether a patient should receive targeted molecular therapy, radiation therapy, and whether they are likely to survive beyond six months. Existing pathway-informed deep learning models have been developed and tested in isolation, making fair comparison across architectures impossible. We present the first unified benchmark for pathway-guided therapy response modeling, evaluating three biologically informed architectures, BINN, GraphPath, and PATH, across five cancer cohorts drawn from The Cancer Genome Atlas, representing 2,622 patients encoded using Reactome pathway activity scores. Each model is trained jointly on all three clinical outcomes under identical data and evaluation conditions, the first study to treat pathway-structured deep learning as a combined therapy and survival prediction problem. Our results show that no single architecture wins across all tasks: PATH performs best for targeted molecular therapy prediction overall, BINN is most reliable for survival prediction, and no model produces useful predictions for radiation therapy, as the key drivers of that decision are clinical variables not captured in gene expression data. Most strikingly, GraphPath achieves an AUROC of 0.92 on prostate targeted molecular therapy prediction, the highest score in the entire benchmark, demonstrating that lateral co-regulation structure produces exceptional discriminative power when matched to a cohort with a narrow targetable driver programme, even under conditions of extreme class imbalance at only 11% positive prevalence.

Keywords

biologically informed neural networks, graph attention, graph transformer, Reactome, ssGSEA, breast cancer, therapy-response prediction, multi-label classification, systems immunology, interpretable deep learning

1 Introduction

Cancer treatment planning is a difficult clinical problem because one patient may need several different kinds of decisions at the same time. A clinician may need to decide whether a patient should receive a targeted molecular therapy, whether radiation therapy should be used, and whether the patient is likely to survive beyond a short-term clinical window. These three questions are related, but they are not the same question. Targeted molecular therapy aims to block specific molecular drivers of cancer growth [29]. Radiation therapy uses ionising radiation to damage tumour cells [2]. Short-term survival prediction is often used to support risk stratification and care planning in oncology [31, 37]. A useful model should therefore learn these outcomes as separate but connected prediction tasks, instead of forcing them into one single label.

Gene expression data can help with this problem because it captures which genes are active in each tumour sample. However, raw gene expression has thousands of genes, and it can be hard to

understand what a model learns from such a large input. A more interpretable approach is to summarize gene expression into biological pathways. Gene set enrichment analysis was introduced to connect gene expression patterns with known biological processes [39], and single-sample GSEA makes it possible to estimate pathway activity for each individual sample [1]. In this paper, we use Reactome, a curated pathway knowledgebase, to convert each patient into a vector of pathway activity scores [11]. This representation is easier to connect back to biology because each feature corresponds to a known pathway rather than an isolated gene.

We formulate our problem as a multi-label prediction task. Given one patient's Reactome pathway activity profile, the model predicts three binary outcomes: targeted molecular therapy (TMT), radiation therapy (RT), and overall survival of at least 180 days ($OS \geq 180$ d). This setup is important because a patient can belong to any combination of these outcomes. For example, one patient may receive targeted therapy but not radiation, while another may receive radiation and also survive beyond 180 days. By using three output heads, the model can learn which pathway patterns are most useful for each clinical question.

Recent work has shown that biological knowledge can be built directly into deep learning models [47, 49, 55]. P-NET used pathway structure to create a neural network for prostate cancer discovery [8]. BINN extended this idea by using biologically informed network connections to support pathway-level interpretation [13, 22]. GraphPath models pathways as nodes in a graph and uses graph attention to learn relationships between pathways [6, 16, 20, 21, 27, 41, 48]. PATH uses a graph transformer [7, 24, 50] to represent pathway interactions for cancer prognosis [14]. These models are promising, but they were developed and evaluated under different settings. Their datasets, labels, input features, and evaluation tasks are not the same, so it is difficult to know whether one architecture is generally better or whether each one works best for different clinical outcomes.

In this work, we compare these pathway-informed models under one common pipeline. We use gene expression and clinical data from The Cancer Genome Atlas (TCGA), accessed through the UCSC Xena platform [12, 44]. We study five solid-tumour cohorts: breast, lung, prostate, head and neck, and thyroid cancer. For every cohort, we use the same preprocessing strategy: gene expression is converted into Reactome pathway activity scores, clinical records are converted into the three binary labels, and the same multi-label learning objective is used for all models. This makes the comparison more direct because the main difference between models is their architecture, not the data pipeline.

Our goal is not only to report which model has the highest score, but also to understand when each kind of biological structure helps. BINN represents Reactome as a sparse hierarchy. GraphPath represents pathways as an attention-based graph. PATH uses a transformer-style graph model with pathway interactions. By testing these approaches on the same task, we ask a simple question:

do different pathway-based architectures help with different therapy and survival outcomes? This question matters because label distributions are different across cancer types. A cohort with many patients receiving targeted therapy may not behave like a cohort where targeted therapy is rare. Therefore, the best model may depend on both the phenotype being predicted and the cancer cohort being studied.

The main contributions of this paper are as follows:

- We build a unified TCGA benchmark for pathway-based prediction of TMT, RT, and $OS \geq 6$ m across five solid-tumour cohorts.
- We use the same Reactome and ssGSEA-based pathway representation for every patient, making the input space consistent across cohorts and models.
- We adapt and compare three pathway-informed deep learning architectures: BINN, GraphPath, and PATH.
- We show that model performance is phenotype-specific, meaning that the strongest architecture can change depending on whether the target is therapy assignment or short-term survival.

2 Data

2.1 Data Curation

We curated a diverse gene expression profiles and clinical phenotype data from the UCSC Xena browser [12], an open-access platform providing harmonised large-scale cancer genomic datasets. Drawing from this resource, we selected five solid-tumour cohorts from The Cancer Genome Atlas [40, 44]: breast, lung, prostate, head and neck, and thyroid cancer. Together, these cohorts encompass a broad spectrum of biologically and clinically distinct tumour types, providing a representative foundation for assessing model generalisability across diverse cancer contexts.

2.2 Task Formation

Building on these cohorts, we defined three binary prediction tasks per patient to support clinically grounded evaluation.

Targeted Molecular Therapy (TMT): Classifying whether a patient received compounds that selectively inhibit molecular drivers of tumour growth (e.g., classifying whether a breast cancer patient received a HER2-directed targeted compound) [29].

Radiation Therapy (RT): Identifying whether a patient underwent ionising radiation treatment designed to induce lethal DNA damage in tumour cells while sparing adjacent healthy tissue (e.g., identifying whether a head and neck cancer patient received radiotherapy as part of their primary treatment course) [2].

Overall Survival ≥ 6 Months ($OS \geq 6$ m): Determining whether a patient survived beyond the clinically established six-month threshold, adopted as a surrogate of short-term prognosis (e.g., determining whether a lung cancer patient remained alive at least 180 days following diagnosis) [31, 37]. We formulate overall survival as a binary task (≥ 6 vs < 6 months) for short-term prognostic stratification in a multi-task setting. This does not model censoring or full time-to-event dynamics, so results reflect short-term risk classification rather than full survival analysis.

2.3 Preprocessing

We aggregated gene expression data into biological pathways using the Reactome database [11], retaining 1,706 curated pathways. Each pathway contains between 10 and 1,000 genes per sample. To quantify the biological activity within each retained pathway, scores were subsequently computed via single-sample gene set enrichment analysis (ssGSEA) [1, 39] and normalised to $[0, 1]$. In parallel, we systematically encoded clinical outcomes as binary labels corresponding to the three prediction tasks defined above: TMT, RT, and $OS \geq 6$ m. Finally, pathway scores and outcome labels were merged using The Cancer Genome Atlas patient identifiers, retaining only complete samples for downstream analysis.

Table 1: Per-Cohort Sample Counts and Positive Prevalence of Each Tumor Phenotype Task Across All Five Cohorts

Cancer Type	<i>n</i>	TMT <i>n</i> (%)	RT <i>n</i> (%)	$OS \geq 6$ m <i>n</i> (%)
Breast	618	571 (92%)	345 (56%)	462 (75%)
Prostate	496	54 (11%)	61 (12%)	480 (97%)
Head & Neck	429	150 (35%)	274 (64%)	395 (92%)
Thyroid	109	7 (6%)	61 (56%)	108 (99%)
Lung	970	311 (32%)	124 (13%)	863 (89%)
Total	2,622	1,093 (42%)	865 (33%)	2,308 (88%)

TMT = targeted molecular therapy; RT = radiation therapy; OS = overall survival.

3 Models

3.1 Model Selection

We evaluate three biologically informed deep learning architectures: **BINN**, **GraphPath**, and **PATH**. Each encodes a progressively richer notion of pathway structure, allowing us to ask how much the structural prior matters for therapy-response prediction. All other experimental conditions are held fixed across models, so any performance differences can be attributed to architectural choice alone.

All three models receive one standardised ssGSEA score per Reactome-matched pathway (1,706 pathways in total) and are trained end-to-end with a multi-task class-weighted binary cross-entropy objective over three prediction heads corresponding to TMT, RT, and $OS \geq 6$ m. The architectures differ only in how biological knowledge is encoded. BINN wires its layers to follow Reactome parent-child relationships exclusively. GraphPath extends this by connecting pathways laterally as well as hierarchically within a graph attention network. PATH goes further by weighting edges with continuous Jaccard similarity and incorporating Laplacian positional encodings and edge-conditioned attention into a Graph Transformer.

3.2 Performance Metrics

We evaluate model performance using five complementary metrics across all benchmark tasks.

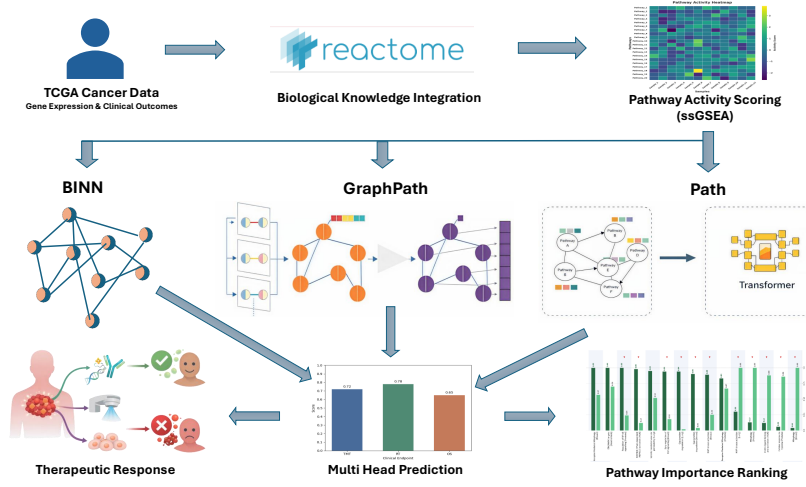


Figure 1: Workflow of the proposed method

AUROC measures the probability that a randomly chosen positive example is ranked higher than a randomly chosen negative:

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (1)$$

AUPRC summarizes the precision-recall trade-off and is particularly informative under class imbalance, which is common in cancer genomics datasets:

$$\text{AUPRC} = \sum_k (R_k - R_{k-1}) P_k \quad (2)$$

F1 Score is the weighted harmonic mean of precision and recall:

$$\text{F1} = \frac{\sum_{i=1}^c 2w_i \cdot \text{precision}_i \cdot \text{recall}_i}{\sum_{i=1}^c w_i (\text{precision}_i + \text{recall}_i)} \quad (3)$$

Accuracy measures the fraction of correctly classified samples:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i] \quad (4)$$

Confusion Matrix provides a full breakdown of predictions across classes, where entry C_{ij} denotes samples of true class i predicted as class j :

$$C = \begin{pmatrix} \text{TP} & \text{FN} \\ \text{FP} & \text{TN} \end{pmatrix} \quad (5)$$

3.3 BINN - Sparse Reactome Hierarchy

Architecture Overview: BINN integrates the Reactome parent-child hierarchy directly into the network by imposing sparse masked-linear connections between layers, together with auxiliary classification heads at each depth. This design ensures that gradient signals reach every intermediate node during training, making gradient-x-input attribution across the hierarchy well-posed [26].

Pathway Hierarchy: Starting from 1,706 Reactome-matched input pathways, the model traverses parent-child edges upward through three hidden layers, yielding a four-layer network with dimensions:

$$1,706 \rightarrow 656 \rightarrow 306 \rightarrow 163. \quad (6)$$

Pathways that terminate before the maximum depth are copied forward, ensuring every pathway is represented at every layer.

Sparse Masked Weights: Between consecutive layers ℓ and $\ell + 1$, the Reactome hierarchy defines a binary connectivity mask $M^{(\ell)} \in \{0, 1\}^{|L_{\ell+1}| \times |L_\ell|}$, where:

$$M_{ij}^{(\ell)} = 1 \iff \text{node } j \text{ in layer } \ell \quad (7)$$

is a Reactome child of node i in layer $\ell + 1$,

or node j is a terminal node copied forward to position i . Weights are initialised with Kaiming uniform initialisation and zeroed outside the mask prior to the first forward pass. Sparsity is enforced throughout training via the elementwise product $W^{(\ell)} \odot M^{(\ell)}$.

Forward Pass: Each hidden block applies the masked linear transformation followed by Tanh activation, Batch Normalisation, and Dropout ($p = 0.2$):

$$\begin{aligned} z^{(\ell)} &= (W^{(\ell)} \odot M^{(\ell)}) h^{(\ell)} + b^{(\ell)}, \\ h^{(\ell+1)} &= \text{Dropout}\left(\text{BN}\left(\tanh\left(z^{(\ell)}\right)\right)\right). \end{aligned} \quad (8)$$

Multi-Layer Supervision: An auxiliary classifier head $\text{Linear}(|L_\ell| \rightarrow 3)$ is attached after every layer, including the raw input ($\ell = 0$), producing three logits per layer. The final predicted probability for head h is the mean of all four sigmoid outputs:

$$\hat{p}_h(x) = \frac{1}{4} \sum_{\ell=0}^3 \sigma\left(H^{(\ell)}\left(h^{(\ell)}\right)[h]\right), \quad (9)$$

where $H^{(\ell)}$ denotes the classifier at layer ℓ . Attaching a loss signal at every depth ensures that each intermediate node receives direct

supervision, a prerequisite for meaningful attribution across the hierarchy.

3.4 GraphPath - Multi-Head Graph Attention

Architecture Overview: GraphPath treats the 1,706 Reactome pathways as nodes in a graph and updates their embeddings through a multi-head graph attention network (GAT). Because pathways interact and co-regulate one another rather than acting as independent units, each pathway aggregates information from its biologically annotated neighbours before the classification step.

Pathway Adjacency Graph: A symmetric binary adjacency matrix $A \in \{0, 1\}^{N \times N}$ is constructed such that:

$$A_{pq} = 1 \iff \text{pathways } p \text{ and } q \text{ share} \\ \text{a Reactome parent-child or sibling relationship.} \quad (10)$$

This yields 4,672 undirected edges with a mean node degree of 5.5. Self-loops are added so that each node attends to its own embedding during message passing.

Input Projection: Each scalar ssGSEA score x_p is projected to a d -dimensional embedding ($d = 64$) via a shared weight matrix $W_{\text{proj}} \in \mathbb{R}^{1 \times d}$ and a pathway-specific bias $b_p \in \mathbb{R}^d$, followed by Tanh activation:

$$h_p^{(0)} = \tanh(x_p \cdot W_{\text{proj}} + b_p). \quad (11)$$

The pathway-specific bias captures baseline differences in pathway activity, while the shared projection weight preserves parameter efficiency.

Multi-Head Graph Attention: Node embeddings are updated by a single GAT layer with $K = 3$ attention heads and ELU output activation. For each head k , the raw attention score and normalised coefficient are:

$$e_{ij}^{(k)} = a^{(k)\top} [W^{(k)} h_i \parallel W^{(k)} h_j], \\ \alpha_{ij}^{(k)} = \text{softmax}_{j \in \mathcal{N}(i)} \text{LeakyReLU}_{0.2}(e_{ij}^{(k)}). \quad (12)$$

Non-edges ($A_{ij} = 0$) are masked to $-\infty$ before the softmax, confining attention to biologically annotated neighbours. Attention dropout ($p = 0.4$) is applied during training. Per-head embeddings are concatenated to yield a node representation of dimension $K \cdot d$.

Readout and Classification: A shared Linear($Kd \rightarrow 1$) layer followed by Tanh collapses each node to a scalar readout. The resulting N -dimensional vector is passed through a fully-connected layer mapping to three binary head logits, followed by sigmoid activation.

3.5 PATH - Edge-Aware Graph Transformer

Architecture Overview: PATH constructs a weighted pathway graph from gene-set overlap and augments a Graph Transformer with Laplacian positional encodings and learnable edge-conditioned attention biases. This allows the model to represent graded biological similarity between pathways and to partially rewire the prior graph when the data warrant it.

Weighted Pathway Adjacency: Edge weights are defined as the Jaccard similarity of Reactome gene memberships:

$$A_{pq} = \frac{|G_p \cap G_q|}{|G_p \cup G_q|}, \quad |G_p| \geq 15, \quad (13)$$

computed from the Reactome GMT file and renormalised to $[0, 1]$. Only pathways with at least 15 annotated genes qualify as nodes, yielding 1,431 nodes, 243,210 weighted edges, and a mean node degree of 339.9. The dense connectivity reflects the substantial gene-set overlap characteristic of the Reactome hierarchy and motivates a soft structural mask rather than hard adjacency truncation.

Input Projection and Positional Encoding: To ensure a fair comparison, we replaced PATH's original Stage 1 (FiLM-modulated gene embeddings) and Stage 2 (intra-pathway attention pooling) with the same per-pathway scalar projection used in GraphPath [34]. This setup uses a shared weight and a pathway-specific bias to project each score into a 64-dimensional space ($\mathbb{R}^d, d = 64$), followed by a Tanh activation.

Laplacian positional encodings are added to endow the model with structural awareness. The top- $k = 16$ non-trivial eigenvectors of the symmetrically normalised Laplacian $L = I - D^{-1/2} A D^{-1/2}$ are concatenated with the normalised node degree $\text{deg}(p)/(N + \epsilon)$, forming a $(k+1)$ -dimensional positional feature per node. These features are projected to $\mathbb{R}^d$ via a learnable linear layer and added to node embeddings before the transformer stack. Eigenvector signs are randomly flipped per epoch during training to resolve sign ambiguity.

Edge-Aware Transformer Blocks. PATH employs $L = 2$ transformer blocks, each computing scaled dot-product multi-head self-attention ($H = 4$ heads) augmented by two structural biases.

Soft structural mask. Non-edges are heavily but softly down-weighted:

$$m_{pq}^{(\text{struct})} = \begin{cases} 0 & \text{if } A_{pq} > 0 \text{ or } p = q, \\ -10 & \text{otherwise.} \end{cases} \quad (14)$$

Keeping non-edges gradient-reachable allows the model to rewire the prior graph when the data provide sufficient evidence.

Edge-conditioned attention bias. A learnable per-layer scalar edge feature is aggregated over attention heads to produce a continuous bias on the attention logits:

$$\phi_{pq}^{(\ell)} = \frac{1}{H} \sum_{h=1}^H \log \text{softplus}(w_h^{(\ell)} e_{pq}^{(\ell)} + b_h^{(\ell)}) + \epsilon, \quad (15)$$

where $e_{pq}^{(\ell)}$ is initialised from the Jaccard weight. The combined bias $m_{pq}^{(\text{struct})} + \phi_{pq}^{(\ell)}$ is added to the raw attention logits before the softmax, grounding attention in biological pathway similarity while permitting data-driven refinement.

Following attention, a residual-wrapped GELU feed-forward network (expansion factor 4) with per-token Batch Normalisation updates node features. A parallel two-layer MLP with Batch Normalisation updates edge features between transformer blocks.

Readout and Classification. Node tokens are collapsed to a graph-level embedding via attention-weighted readout:

$$g = \sum_p w_p x_p^{(L)}, \quad (16)$$

$$w_p = \text{softmax}_p \left(v^\top \tanh \left(U x_p^{(L)} \right) \right),$$

where $U \in \mathbb{R}^{d \times (d/2)}$ and $v \in \mathbb{R}^{d/2}$ are learnable parameters. This soft attention over nodes identifies the pathways most informative for the three prediction targets. The graph embedding g is then passed through a classification head:

$$\begin{aligned} \text{Linear}(d \rightarrow d) &\rightarrow \text{BN} \rightarrow \text{GELU} \\ &\rightarrow \text{Dropout}(p = 0.2) \rightarrow \text{Linear}(d \rightarrow 3), \end{aligned} \quad (17)$$

producing three binary head logits.

3.6 Loss, Training, and Reproducibility

Objective Function: All three models minimise a per-head class-weighted binary cross-entropy loss. Positive-class weights are computed on the training fold as $w_h = \text{neg}_h / \text{pos}_h$ and clipped to $[0.1, 20]$ to prevent extreme imbalance from destabilising training. For BINN, the loss is applied to the averaged sigmoid outputs; for GraphPath and PATH, it is applied to raw logits via BCEWithLogitsLoss (or `F.binary_cross_entropy_with_logits`) for numerical stability.

Optimisers and Learning Rate Schedules: Optimiser choices follow each reference implementation:

Table 2: Optimiser configuration per model.

	BINN	GraphPath	PATH
Optimiser	Adam [17]	SGD (momentum 0.9)	AdamW [25]
LR	10^{-3}	5×10^{-2}	10^{-4}
Weight Decay	10^{-3}	5×10^{-2}	5×10^{-4}
Batch Size	32	32	16

All three models employ ReduceLROnPlateau scheduling on validation loss with a reduction factor of 0.5 and plateau patience of 7, 8, and 10 epochs for BINN, GraphPath, and PATH respectively.

Early Stopping: Training is terminated when validation loss fails to improve for 20 consecutive epochs (BINN) or 25 consecutive epochs (GraphPath, PATH). For PATH, a minimum of 25 training epochs is required before the patience counter activates, preventing premature termination during the characteristically slow initial convergence of the transformer stack. The checkpoint with the lowest validation loss is restored prior to evaluation.

Reproducibility: All models are implemented in PyTorch 2.0 [32], with scikit-learn [33] used for metric computation. Random seeds are fixed identically across Python, NumPy, and PyTorch (seed = 42) for all models and cohorts. All figures reported in this study are regenerated by executing `scripts/run_all.sh` within the `binnn/`, `graphpath/`, and `path/` directories, followed by `paper/build.sh`.

4 Results

Performance was evaluated across three clinical prediction heads TMT, RT, and OS using F1, AUPRC, and AUROC metrics on five TCGA solid-tumour cohorts. The nine-panel benchmark presented in Fig. 2 makes one thing immediately clear: no single architecture dominates uniformly. Direct comparison with the originally reported results of BINN, GraphPath and PATH is not appropriate because the models were evaluated under a different prediction task, feature representation, cohort composition and endpoint definition.

4.1 Comparative Performance Across Clinical Endpoints

4.1.1 Evaluation of BINN: BINN is the most stable of the three architectures, most apparent on the OS head (Fig. 2). It leads on both Breast and Lung OS by AUROC, with Lung returning a strong OS AUPRC of 0.89, though GraphPath records the highest Lung OS AUPRC in the benchmark at 0.93. This consistency is driven by multi-depth auxiliary supervision that propagates loss signals through every layer of the Reactome hierarchy, reinforcing broad transcriptional survival signals distributed across multiple pathway levels. The Thyroid OS-F1 of 1.00 is nominally the strongest single figure in the benchmark, though the small cohort size ($n=109$) and near-universal OS prevalence (99%) limit the reliability of this result.

On the TMT head, BINN shows strong class balance on Breast cohorts but fails to maintain competitive discriminative power, yielding the lowest TMT-AUROC among the three architectures (0.60). The strict parent-child wiring restricts lateral information flow between pathways, limiting discrimination of near-threshold TMT cases. On RT, BINN leads on Breast and Prostate, most notably achieving a Prostate RT-AUROC of 0.85, but trails GraphPath on Head & Neck by both F1 and AUPRC. Head & Neck RT appears driven more by lateral co-regulatory interactions than by vertical hierarchy alone, which the strict hierarchical structure of BINN cannot adequately capture.

4.1.2 Evaluation of Path: PATH achieves the highest TMT-AUROC on Breast (0.64), with its advantage widening further on Head & Neck (0.73), where it leads BINN by eight points and GraphPath by a wide margin (Fig. 2). The Jaccard-weighted adjacency assigns continuous similarity scores between pathways, and this graded representation of co-regulatory overlap appears well-suited to Head & Neck’s targetable driver landscape. It is an advantage that GraphPath’s binary adjacency cannot replicate.

On the RT head, all three models converge on Breast RT, where the near-balanced label distribution (56%) reduces the discriminative signal and makes architectural differences largely irrelevant. PATH records the weakest Prostate RT results of the three, as its dense graph structure aggregates signals from pathways unrelated to the narrow RT-relevant programme in this cohort. On OS, PATH leads only on Prostate (AUROC 0.65), where continuous edge weighting detects subtle survival-associated transcriptional differences that both competing architectures fail to resolve.

4.1.3 Evaluation of GraphPath: GraphPath shows the widest performance range of the three models, visible across both the AUROC and AUPRC rows of Fig. 2. Its most striking result is a Prostate TMT-AUROC of 0.92, the highest in the entire benchmark. This

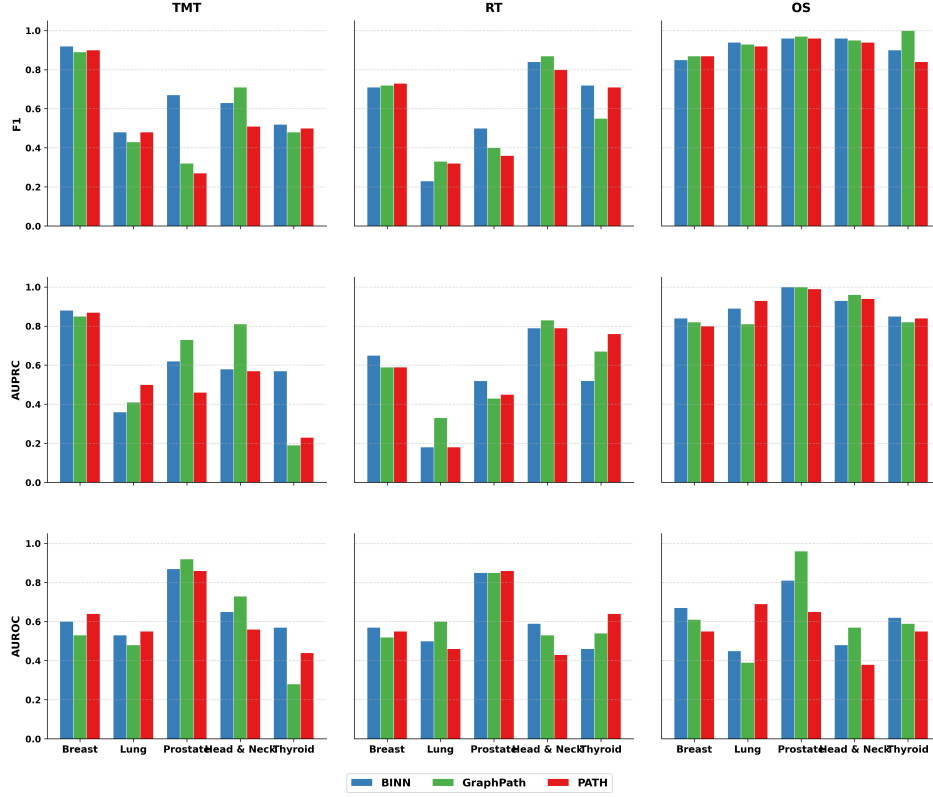


Figure 2: Held-out test-set F1, AUPRC, and AUROC per clinical head (TMT, RT, OS) across all five TCGA cohorts. BINN blue, GraphPath green, PATH red [30, 46].

performance is particularly notable given the cohort’s low TMT prevalence of only 11%, suggesting that lateral co-regulation structure is especially well-suited for identifying the narrow targetable driver programme present in this cohort. On RT, GraphPath leads Head & Neck across all three metrics, where the high RT label prevalence (64%) provides sufficient representation for neighbourhood aggregation to be effective.

On OS, GraphPath is competitive on Lung AUPRC (0.93) and Head & Neck F1 (0.95), but records a Prostate OS-AUROC of 0.39, the lowest in the entire evaluation. The same model that leads Prostate TMT thus fails on Prostate OS, indicating that sibling-edge adjacency effective for therapy targeting does not extend to survival stratification in this cohort.

4.2 Cross-Cancer Subtype Performance

As shown in Fig. 3, performance variability across cohorts is more head-dependent than model-dependent. On the TMT head, the three models behave quite differently across cancer types. GraphPath performs strongest on Prostate, where lateral co-regulation structure is well-matched to the narrow targetable driver programme in this cohort. GraphPath drops sharply on Head & Neck, producing the most uneven profile among all models - consistent with binary adjacency being insufficient to capture the overlapping pathway activity that drives TMT response in this cancer type [27]. PATH

shows a more even spread across cohorts, with its strongest performance on Head & Neck, where graded edge weighting appears most relevant [14].

On the RT head, all three models perform similarly across cohorts. The clearest separation appears at Head & Neck, where GraphPath performs best, and at Thyroid, where PATH holds a slight advantage. On Breast and Lung, all three models record near-identical performance, confirming that the RT signal in these cohorts is not strong enough for architectural differences to have any meaningful effect. On the OS head, all three models track closely across most cohorts. The only notable difference is at Prostate, where PATH outperforms both competitors, and at Head & Neck, where BINN falls slightly behind. Across all three heads, TMT shows the widest performance spread between models and OS the narrowest, suggesting that therapy-specific outcomes are more sensitive to architectural choice than survival prognosis.

4.3 Prediction Reliability Assessment

BINN was evaluated on a 70/15/15 split (396 test samples) and GraphPath and PATH on an 80/10/10 split (263 test samples), following each model’s original reference implementation. On the TMT head, all three models show a reasonable ability to detect true positive cases, with BINN recording the highest TP count. Low FN counts across all models indicate that pathway-level features

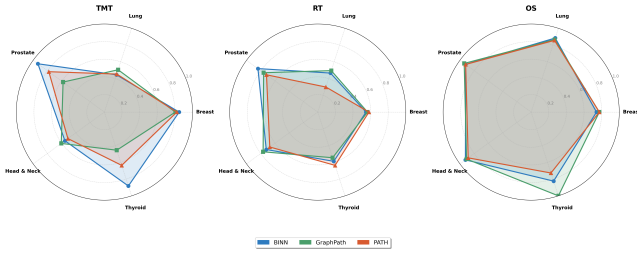


Figure 3: Pan-cancer AUROC per (model, cohort) cell for each clinical head. Dotted line marks chance level.

encoding targetable molecular drivers are sufficiently captured [29]. The elevated FP counts reflect the difficulty of separating the small TMT-negative subgroup from the dominant positive class in a 92% prevalence cohort, rather than a systematic prediction failure.

On the RT head, all three models maintain consistent TP rates despite the near-balanced label distribution, which is the most demanding setting for threshold-based prediction. BINN and GraphPath show comparable FP behaviour, while PATH’s higher FP reflects its tendency to capture broader pathway co-activation patterns beyond the RT-specific signal. FN counts remain low across all models, indicating that true RT-positive cases are well-represented in the learned pathway features.

On the OS head, all three models perform most reliably. GraphPath leads with the highest TP rate, followed closely by BINN and PATH, with FP counts remaining modest and consistent. FN counts are the lowest across all three heads, indicating that survival-associated pathway signatures are the most consistently learned biological signal across all architectures [15, 19, 23, 28, 35, 42, 43, 45, 51, 54]. The full prediction breakdown is summarised in Fig. 4.

	BINN		GraphPath		PATH	
TMT	21 FN (5.3%)	118 TN (29.8%)	22 FN (4.6%)	65 TN (26.7%)	28 FN (10.4%)	75 TN (28.5%)
	147 TP (37.3%)	110 FP (27.8%)	94 TP (35.7%)	82 FP (31.2%)	88 TP (33.5%)	72 FP (27.4%)
RT	20 FN (5.1%)	115 TN (29.8%)	13 FN (4.9%)	73 TN (27.8%)	12 FN (4.4%)	49 TN (18.6%)
	124 TP (31.3%)	137 FP (34.0%)	87 TP (33.1%)	90 FP (34.2%)	88 TP (33.5%)	114 FP (43.3%)
OS	13 FN (3.3%)	3 TN (0.8%)	5 FN (1.9%)	2 TN (0.8%)	11 FN (4.2%)	2 TN (0.8%)
	338 TP (85.4%)	42 FP (10.8%)	227 TP (86.3%)	29 FP (11.0%)	221 TP (84.0%)	29 FP (11.0%)

Legend: TP - True Positive, TN - True Negative, FP - False Positive, FN - False Negative

Figure 4: Confusion-matrix counts on the test split at the 0.5 threshold. Each row a model, each column a clinical head.

4.4 Pathway Importance Analysis and Biological Interpretation

Normalized pathway importance scores from BINN and GraphPath were compared across the TMT, RT, and OS heads to assess whether the two architectures converged on similar biological mechanisms. PATH was excluded because it does not generate pathway-level importance scores. Pathways with score differences below 0.50 were considered convergent and prioritized for biological validation, whereas larger differences (>0.50 ; red arrows in Fig. 5) were interpreted as potential architecture-specific effects.

For the TMT head, the models emphasized different layers of the same oncogenic process, with BINN highlighting upstream transcriptional and inflammatory regulation and GraphPath focusing on downstream translational maintenance. Despite these differences, both converged on **ERK/MAPK Targets** [38] in Lung cancer, suggesting a shared biologically meaningful signal.

For the RT head, the strongest convergence was observed for **Receptor-Mediated Mitophagy** [53] [18] and **ERK/MAPK Targets** [3]. Both pathways have established links to radioresistance through mechanisms involving cellular stress adaptation, DNA repair, and survival signaling. Additional convergence on **VEGFR2-Mediated Vascular Permeability** [10] further supports the role of tumour microenvironment and vascular signaling in radiation response.

The OS head showed the highest agreement between the two models. Both architectures consistently ranked **ERK/MAPK Targets** [52] [9] among the most important pathways in both Lung and Head & Neck cohorts, making it the strongest convergent signal identified in this study. Its consistent appearance across multiple cohorts and clinical endpoints provides strong support for its biological relevance. In contrast, pathways related to apoptosis and metabolism showed lower agreement between the models, suggesting architecture-dependent interpretations that warrant further investigation.

5 Discussion

This benchmark highlights that model performance is highly task-dependent, with no single architecture consistently outperforming the others across all clinical heads. PATH achieved the strongest performance on the TMT task, suggesting that modeling pathways with weighted connections helps preserve biologically meaningful relationships between interconnected processes and therapy-associated molecular signals. In contrast, BINN showed the most consistent performance for OS prediction, indicating that its hierarchical structure is effective at capturing survival-related biological signals across different levels of pathway organization. For RT prediction, all architectures achieved comparable results, suggesting that the main limitation may lie in the available features rather than the model design itself. This finding implies that improving RT prediction will likely require integrating additional clinical and molecular information beyond transcriptomic pathway features alone. The pathway analysis further highlighted the importance of interpretable model architectures. Both BINN and GraphPath consistently identified the ERK/MAPK Targets pathway as highly influential across multiple cohorts and clinical outcomes, increasing confidence that this signal reflects a genuine biological mechanism.

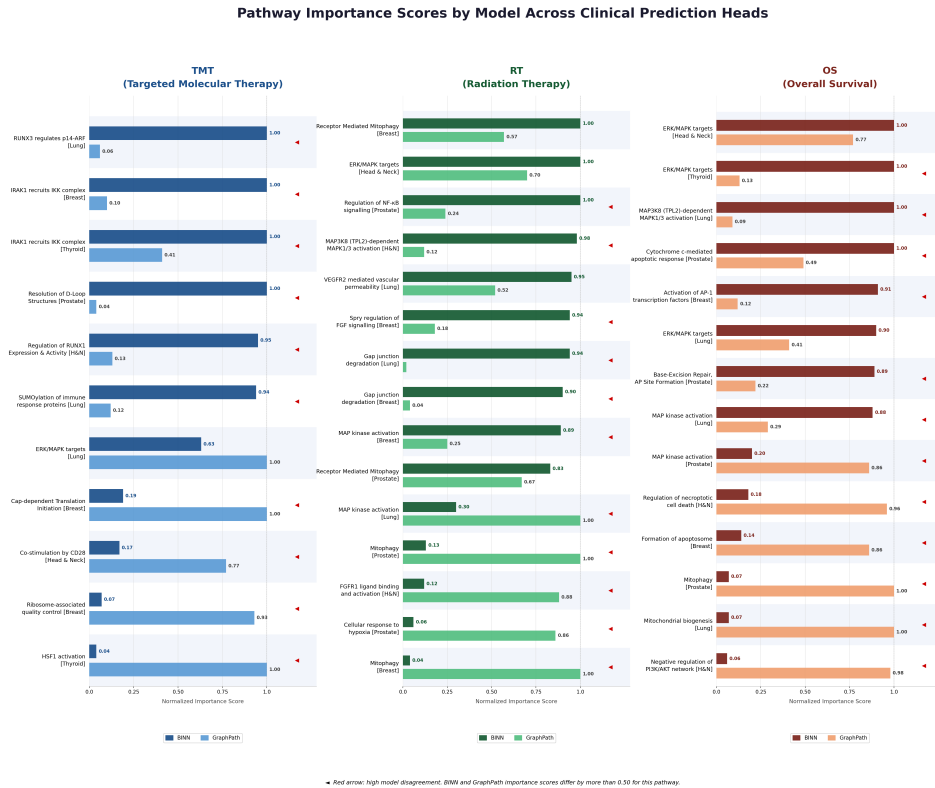


Figure 5: Normalised pathway importance scores for BINN and GraphPath across all three clinical heads. Red arrows mark pathways with inter-model disagreement > 0.50 .

rather than a model-specific effect [5]. While PATH achieved strong predictive performance, its lack of pathway-level interpretability makes it more difficult to understand the biological basis of its predictions, which may limit its usefulness for mechanistic insights and future clinical applications [4, 36]. Several limitations should be acknowledged. TCGA records treatment exposure rather than true response, and some cohorts have few positive cases, which may reduce result stability. Thus, findings should be viewed as indicative until validated on larger response-labelled datasets. Overall, results suggest that architecture choice should be task-specific: PATH is best suited for TMT prediction, BINN for OS prediction, and RT prediction likely requires multimodal molecular and clinical data. Additionally, PATH uses a gene-membership filter, making the benchmark not fully input-consistent; future work should standardize pathway selection.

6 Conclusion

This study provides a unified benchmark for pathway-informed deep learning in cancer therapy and prognosis prediction. Using Reactome-derived pathway activity profiles from TCGA transcriptomic data, we evaluated BINN, GraphPath, and PATH across five solid-tumor cohorts and three clinically relevant outcomes: targeted molecular therapy (TMT), radiation therapy (RT), and overall survival ($OS \geq 6$ months).

The results demonstrate that pathway-guided architectures can effectively leverage biological knowledge for clinical prediction while providing interpretable pathway-level insights. However, no single model consistently outperformed the others across all cancer cohorts and prediction tasks, highlighting that model performance depends on both the clinical endpoint and the underlying disease context. These findings suggest that the choice of pathway-informed architecture should be guided by the specific prediction objective rather than by aggregate benchmark performance alone.

Beyond predictive accuracy, this work establishes a standardized and reproducible evaluation framework for comparing pathway-based deep learning models under consistent data, preprocessing, training, and evaluation settings. By bringing multiple clinically relevant prediction tasks into a single benchmark, the study provides a foundation for future research in biologically informed machine learning and precision oncology. Future work should explore response-labeled cohorts, integrate multimodal clinical and molecular data, and validate findings in larger independent datasets to further improve the clinical utility of pathway-guided predictive models.

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